

Special Relativity: Electromagnetic Waves and the Maxwell Action

Based on lectures by Leonard Susskind

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Chapter 1

Electromagnetic Waves and the Maxwell Action

Overview. We first solve the source-free Maxwell equations for a plane wave and let the equations determine its polarization, field geometry, and speed. We then ask a deeper question: which local, Lorentz-invariant, gauge-invariant action produces the dynamical Maxwell equations? The answer ties the electromagnetic field, charge conservation, and gauge invariance into one structure.

1.1 Two Families of Maxwell Equations

We begin with a plane electromagnetic wave. Here *plane* is the geometrical word, not *plain* in the sense of unadorned. The immediate questions are concrete: what does such a wave look like, and how does its form follow from Maxwell's equations?

It is useful to put the equations into two columns before doing anything with the wave. In empty space they are

$$\nabla \cdot \mathbf{B} = 0, \quad \dot{\mathbf{B}} = \nabla \times \mathbf{E}, \quad (1.1)$$

$$\nabla \cdot \mathbf{E} = 0, \quad \dot{\mathbf{E}} = -\nabla \times \mathbf{B}. \quad (1.2)$$

We use the mostly-plus metric

$$\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1) \quad (1.3)$$

and the course convention

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad F_{0i} = E_i, \quad F_{ij} = \epsilon_{ijk} B_k. \quad (1.4)$$

Then the same two families can be written

$$\partial_\mu F_{\nu\sigma} + \partial_\nu F_{\sigma\mu} + \partial_\sigma F_{\mu\nu} = 0, \quad (1.5)$$

$$\partial_\mu F^{\mu\nu} = 0. \quad (1.6)$$

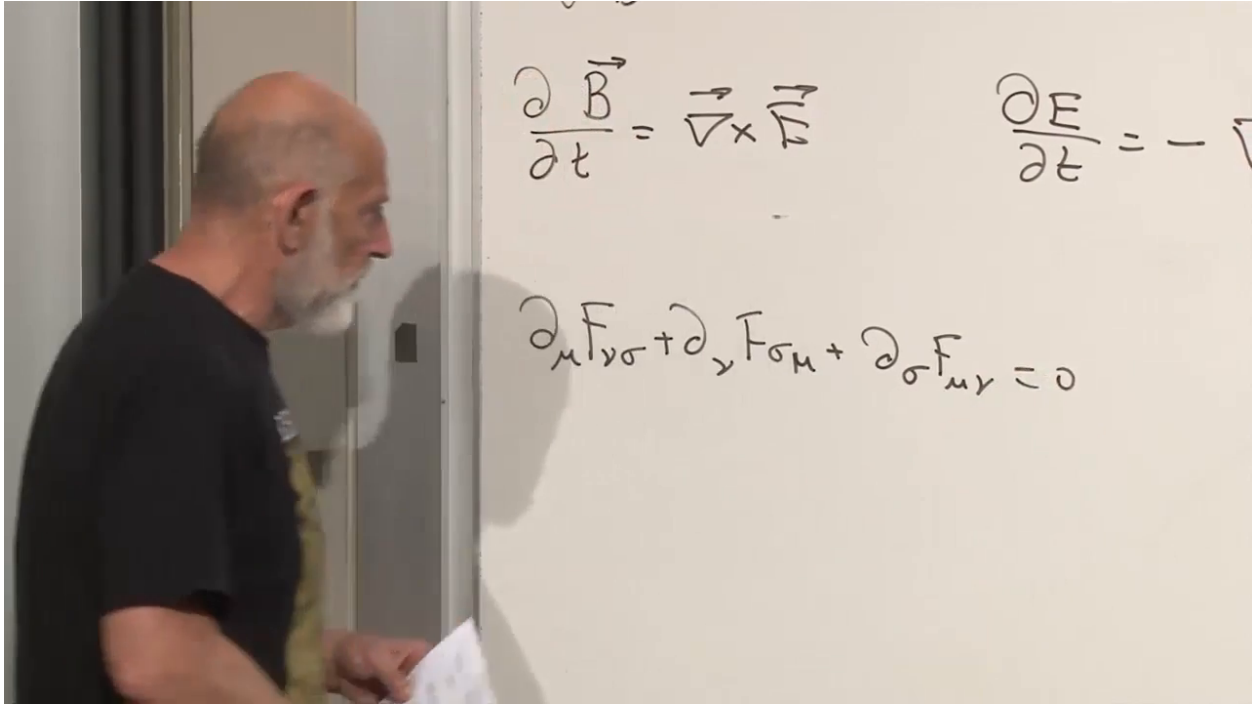


Figure 1.1: The two vacuum Maxwell families and their covariant forms. The cyclic equation is an identity; the contracted equation is dynamical.

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The distinction between the two rows matters. Equation (1.5) follows identically from the existence of the potential A_μ : every term is a second derivative, and the terms cancel in pairs. It is therefore a Bianchi identity. Equation (1.6), by contrast, is an equation of motion. Change the electromagnetic Lagrangian and this is the family that changes.

There is a small observation on the board that will do two jobs in this chapter. Every field in Maxwell's equations appears differentiated. That fact will constrain the relative phases of a plane wave, and it will later make gauge invariance natural.

1.2 A Plane Wave Along the z -Axis

Choose coordinates so that the wave propagates along the z -axis. This costs no generality; a wave in another direction can be brought to this form by rotating the axes. Let k be its wave number,

$$\lambda = \frac{2\pi}{k}, \quad (1.7)$$

and let ω be its frequency. Before imposing any equation of motion, write all six field components with a common phase

$$\theta = kz - \omega t : \quad (1.8)$$

$$\begin{aligned} E_x(z, t) &= \epsilon_x \sin \theta, & B_x(z, t) &= \beta_x \sin \theta, \\ E_y(z, t) &= \epsilon_y \sin \theta, & B_y(z, t) &= \beta_y \sin \theta, \\ E_z(z, t) &= \epsilon_z \sin \theta, & B_z(z, t) &= \beta_z \sin \theta. \end{aligned} \quad (1.9)$$

The six Greek letters are constant amplitudes.

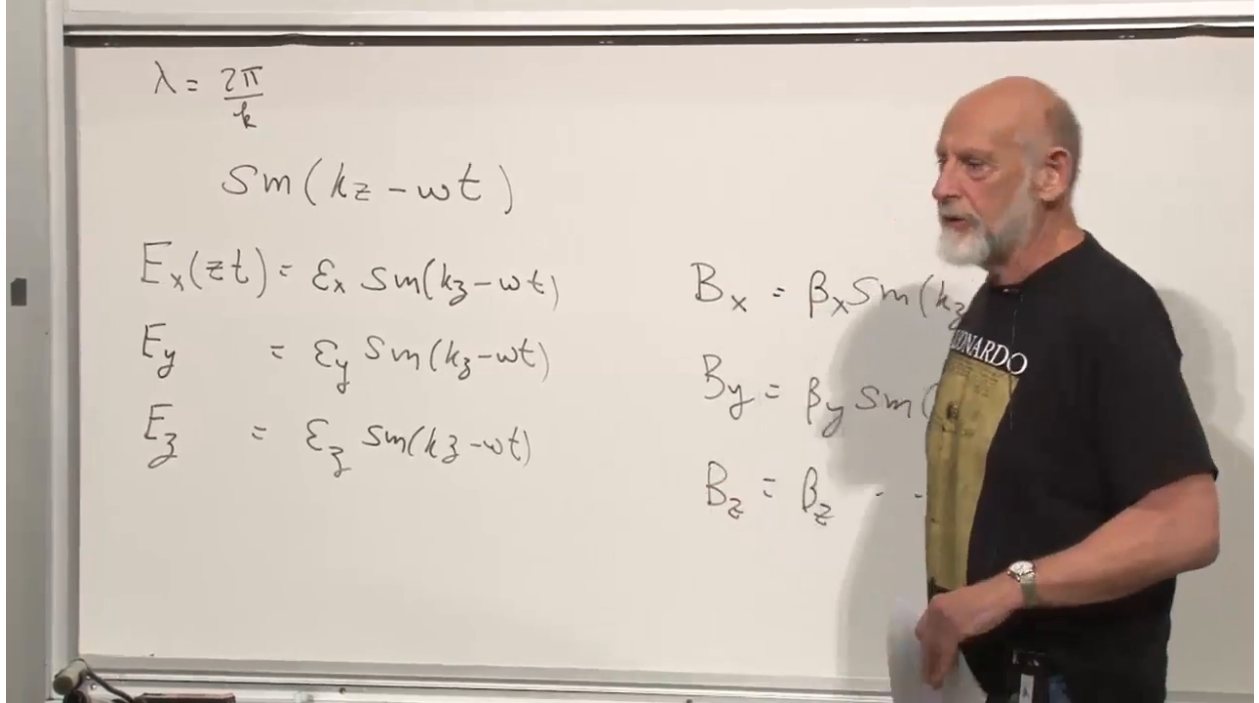


Figure 1.2: The six-component plane-wave ansatz before Maxwell's equations remove the forbidden amplitudes.

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Why begin with the same sinusoidal dependence for $\mathbf{E}$ and $\mathbf{B}$? Suppose one field carried $\sin \theta$ and the other carried $\cos \theta$. A time derivative on one side of a curl equation and a spatial derivative on the other would then produce different trigonometric functions. The equality could not hold for every z and t . A common constant phase is allowed, but translating the origin of z removes it. Thus the ansatz is not an extra dynamical law; it is the simplest representative of the phase relation the differential equations will require.

No Maxwell equation has yet been used. We have only chosen a direction, a wavelength, and a convenient phase origin. The equations will now decide which amplitudes survive.

1.2.1 The divergence constraints

Because the fields depend only on z and t ,

$$\nabla \cdot \mathbf{E} = \partial_x E_x + \partial_y E_y + \partial_z E_z = \partial_z E_z. \quad (1.10)$$

In empty space this must vanish for every phase, so

$$\epsilon_z = 0. \quad (1.11)$$

The identical argument for the magnetic field gives

$$\beta_z = 0. \quad (1.12)$$

Neither field has a component along the direction of propagation. The wave is transverse.

There remains a rotational freedom in the transverse xy -plane. The pair (ϵ_x, ϵ_y) is an ordinary two-dimensional vector, so rotate the transverse axes until

$$\epsilon_y = 0, \quad \epsilon_x \neq 0. \quad (1.13)$$

This is a choice of linear polarization, not a restriction on the physics.

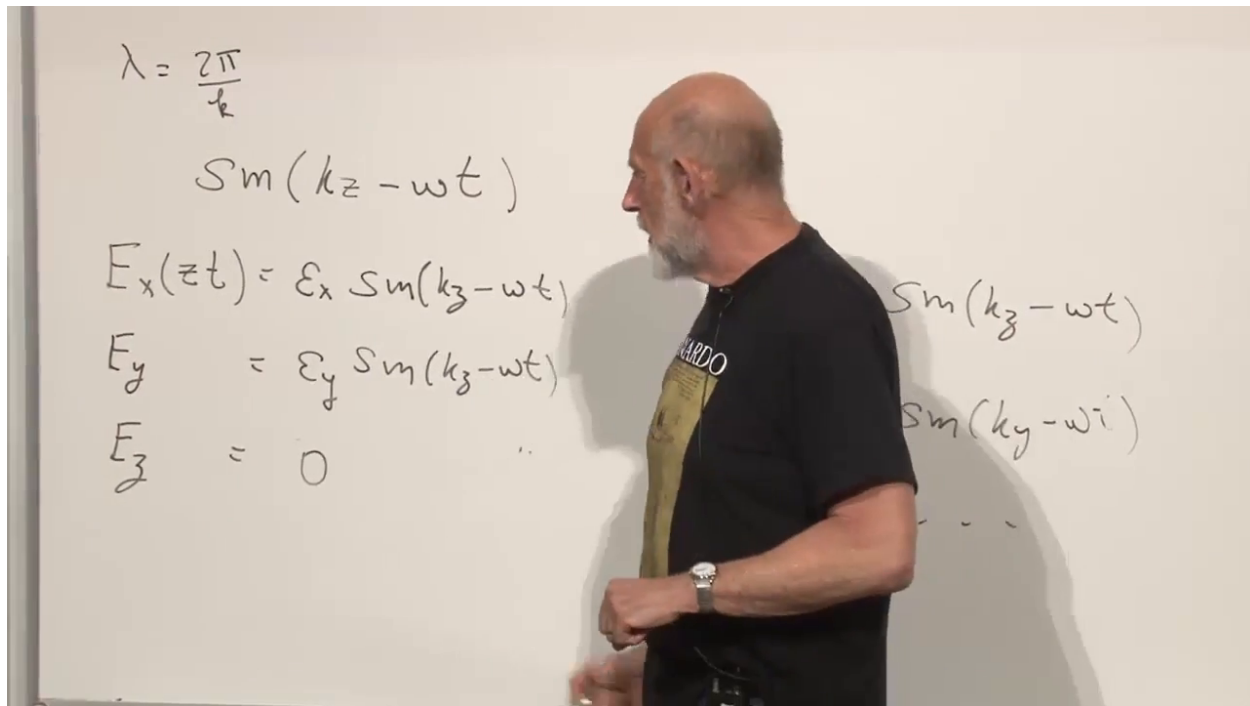


Figure 1.3: The longitudinal amplitudes have vanished and the transverse electric axis has been chosen.

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1.2.2 The curl equations

Use the x -component of $\dot{\mathbf{E}} = -\nabla \times \mathbf{B}$. Since the fields vary only with z ,

$$\dot{E}_x = -(\nabla \times \mathbf{B})_x = \partial_z B_y. \quad (1.14)$$

Substituting the ansatz gives

$$\begin{aligned} -\omega \epsilon_x \cos \theta &= k \beta_y \cos \theta, \\ \beta_y &= -\epsilon_x \frac{\omega}{k}. \end{aligned} \quad (1.15)$$

The y -component of the same Maxwell equation reads

$$\dot{E}_y = -(\partial_z B_x). \quad (1.16)$$

The left side is zero by our polarization choice, so

$$\beta_x = 0. \quad (1.17)$$

$$-\frac{\partial E_y}{\partial t} = -(\nabla \times \mathbf{B})_y$$

$$\frac{\partial E_y}{\partial t} = \partial_x B_z - \partial_z B_x$$

$$-E_x \omega = \beta_y k$$

Figure 1.4: A component curl calculation relates ϵ_x to β_y and removes the remaining magnetic amplitude.

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The geometry is therefore fixed:

$$\mathbf{E} \parallel \hat{\mathbf{x}}, \quad \mathbf{B} \parallel \hat{\mathbf{y}}, \quad \mathbf{k} \parallel \hat{\mathbf{z}}. \quad (1.18)$$

The electric field, magnetic field, and direction of travel are mutually perpendicular.

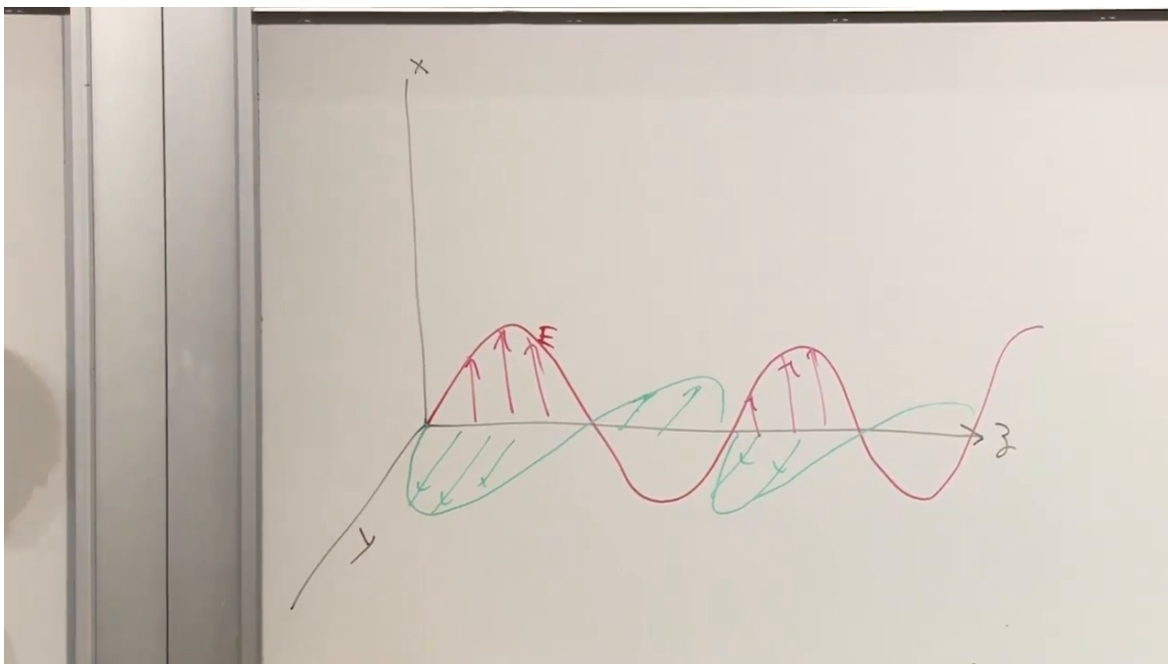


Figure 1.5: The board picture of the transverse electric and magnetic oscillations advancing along the z -axis.

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The other curl equation supplies an independent amplitude relation. Its y -component is

$$\dot{B}_y = (\nabla \times \mathbf{E})_y = \partial_z E_x, \quad (1.19)$$

and therefore

$$\begin{aligned} -\omega \beta_y \cos \theta &= k \epsilon_x \cos \theta, \\ \epsilon_x &= -\beta_y \frac{\omega}{k}. \end{aligned} \quad (1.20)$$

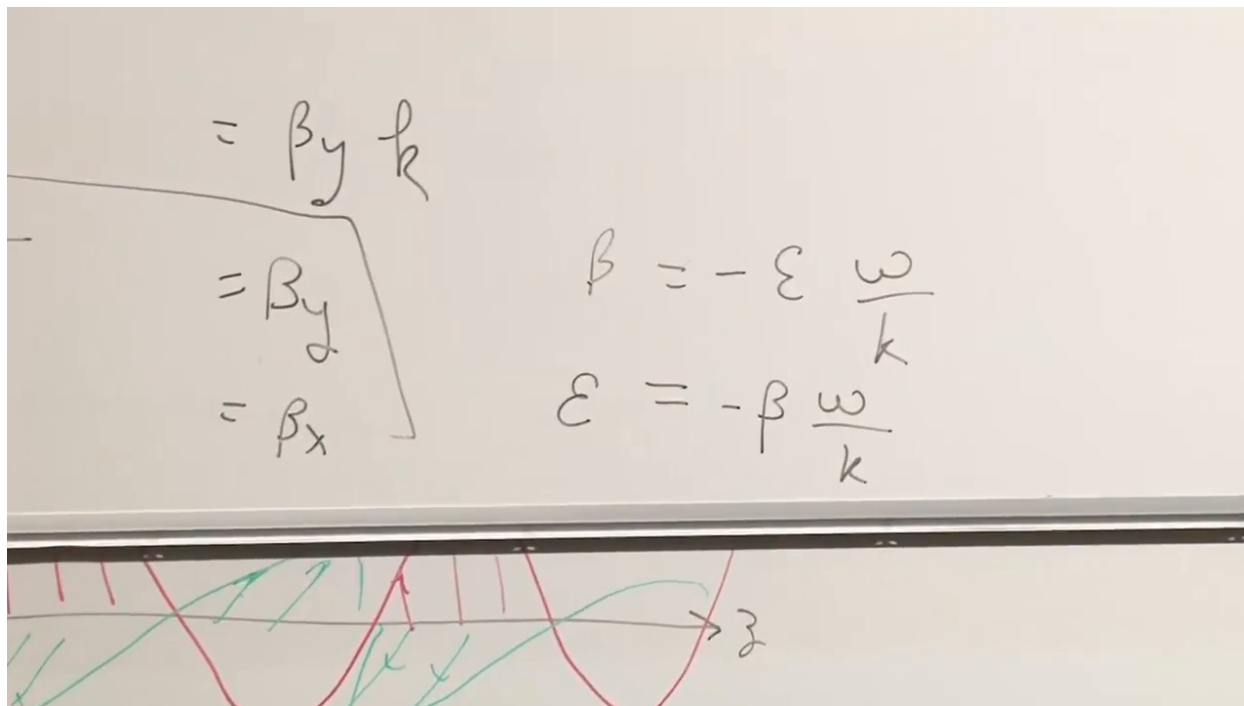


Figure 1.6: The reciprocal amplitude relations above the transverse-wave sketch.

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Combining equations (1.15) and (1.20) gives the dispersion relation

$$\left(\frac{\omega}{k}\right)^2 = 1. \quad (1.21)$$

For the positive-frequency wave moving in the chosen direction,

$$\omega = k \quad (c = 1). \quad (1.22)$$

The common phase is consequently

$$\theta = k(z - t). \quad (1.23)$$

Restoring ordinary units gives

$$\omega = ck, \quad \theta = k(z - ct). \quad (1.24)$$

Thus the profile moves rigidly at the speed of light. In the units used on the board, the electric and magnetic amplitudes have equal magnitude. Their relative minus sign is an orientation convention: it says which way the chosen y -axis points, not that one field lags the other by a quarter cycle.

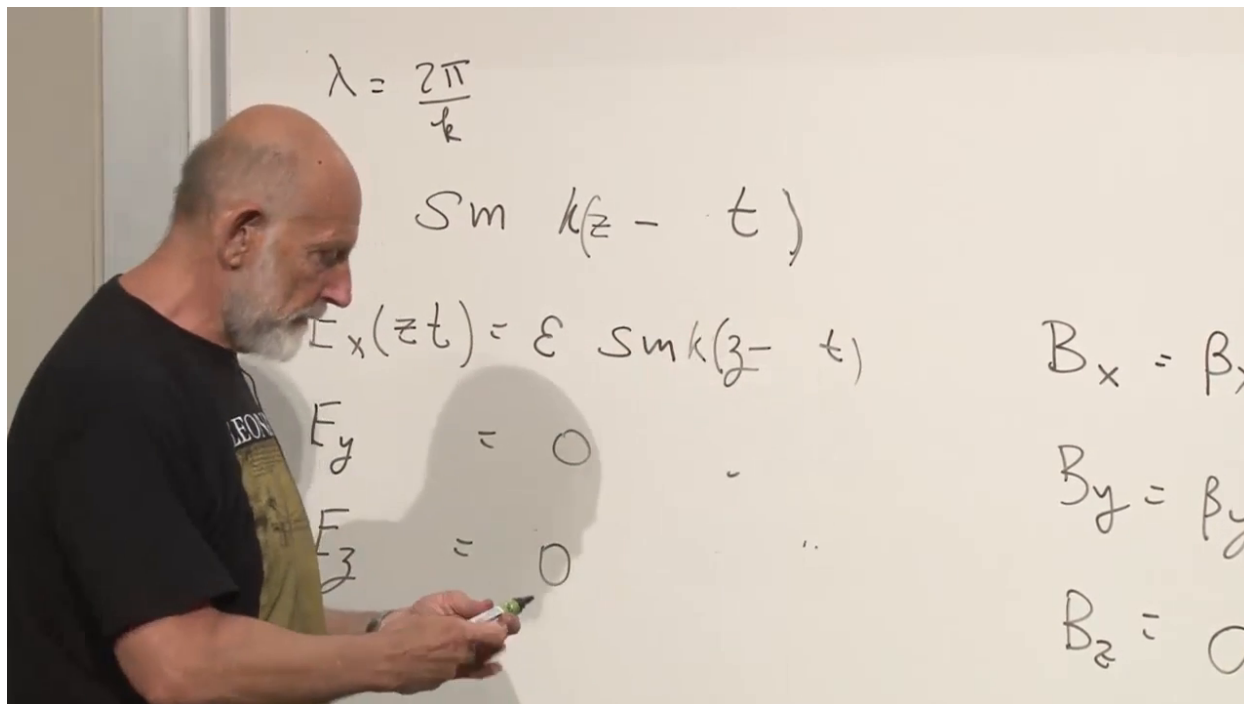


Figure 1.7: The final plane-wave form after $\omega = k$ has been imposed.

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Question from the audience. Do ω and k not have different units?

Response. They do in ordinary units: ω is inverse time and k is inverse length. With $c = 1$, time and length have the same unit. Restoring c changes $\omega = k$ into $\omega = ck$, exactly as dimensional analysis requires. ^a

^aLecture timestamp: 00:25:55.

Question from the audience. Are the electric and magnetic fields always in phase in a plane wave?

Response. They have the same phase dependence. Their signed amplitudes can differ by a minus sign, which may be described as a 180° reversal, but Maxwell's equations do not permit a 90° sine-cosine lag for this linearly polarized plane wave. Circular polarization is assembled from transverse components and requires more careful language, but it does not alter this componentwise Maxwell relation. ^a

^aLecture timestamp: 00:27:31.

Question from the audience. If both fields pass through zero together, is energy still conserved?

Response. Yes. The energy density at one fixed point oscillates as the wave passes, but the energy has moved rather than disappeared. What occupied one region at one time occupies a neighboring region a moment later. Local energy transport, represented more fully by the Poynting flux, reconciles the changing density with conservation. ^a

^aLecture timestamp: 00:28:15.

Maxwell's equations are linear. The amplitude ϵ_x may be large or small, β_y follows from it, and any sum or difference of solutions is again a solution. Waves traveling in different directions may therefore be superposed. This brief calculation is worth doing: an account of electromagnetism should not leave the electromagnetic wave as an unexplained quotation.

Question from the audience. Did the derivation assume from the start that $\mathbf{E}$ and $\mathbf{B}$ had the same phase?

Response. The common phase was written as an ansatz, but it is not an independent physical assumption. Because each side of the curl equations differentiates a field, the equations eliminate a nontrivial relative phase. A common phase shift can then be removed by translating the origin.^a

^aLecture timestamp: 00:29:56.

Question from the audience. Where did the second relation $\epsilon_x = -\beta_y \omega/k$ come from?

Response. The first relation came from $\dot{\mathbf{E}} = -\nabla \times \mathbf{B}$. The second comes from the companion equation $\dot{\mathbf{B}} = \nabla \times \mathbf{E}$, with $\mathbf{E}$ and $\mathbf{B}$ interchanged in the same component calculation. Both are needed to obtain $\omega^2 = k^2$.^a

^aLecture timestamp: 00:30:25.

1.3 Why Differential Equations Are Not Enough

We now turn from a particular solution to the structure of the theory. The principle of least action is what organizes energy conservation, momentum conservation, and their relation to spacetime symmetries. One can invent differential equations that possess no energy worth calling conserved and no useful symmetry principle. Maxwell's equations should instead arise from a Lagrangian.

Only the dynamical family needs to be produced:

$$\partial_\mu F^{\mu\nu} = J^\nu. \quad (1.25)$$

The Bianchi family already follows from the definition of $F_{\mu\nu}$. In three-vector language, our convention makes

$$\nabla \cdot \mathbf{E} = \rho, \quad (1.26)$$

$$\dot{\mathbf{E}} + \nabla \times \mathbf{B} = -\mathbf{J}. \quad (1.27)$$

The minus sign in (1.27) is important. It is the component form of (1.25) under (1.4), and it will be rediscovered in a live sign check near the end.

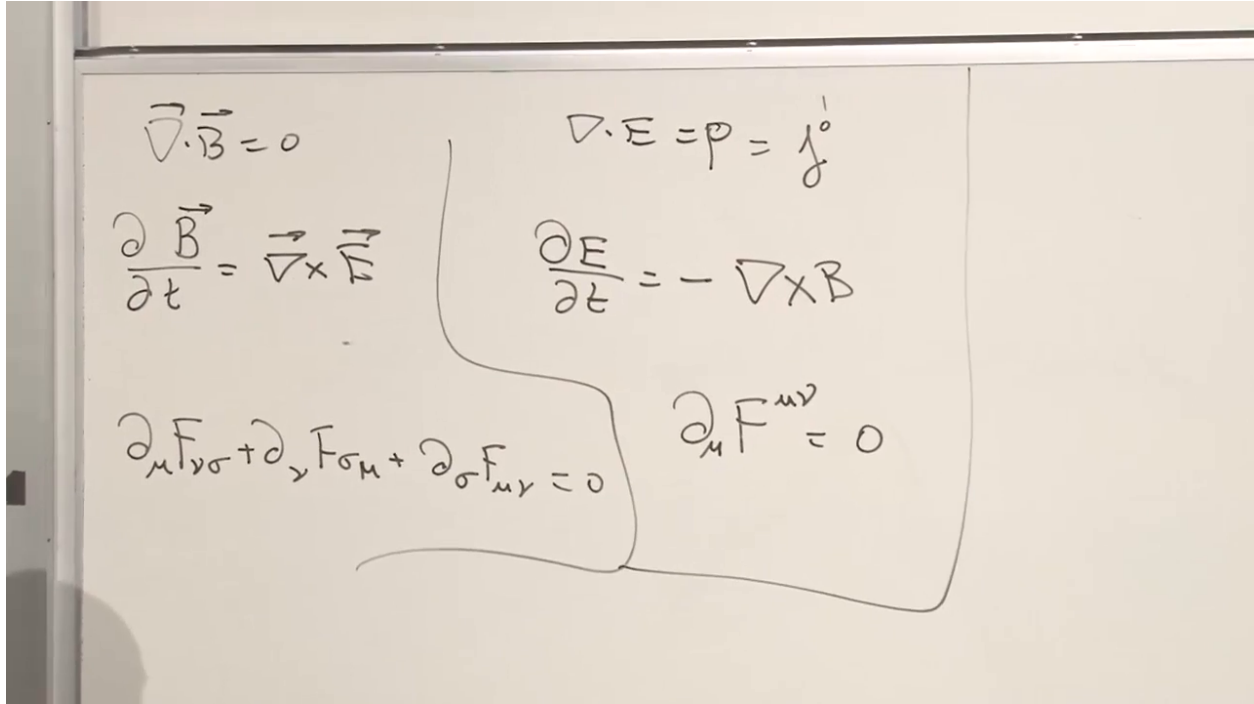


Figure 1.8: The full Maxwell board at the transition from the wave solution to the action principle.

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1.4 Local Relativistic Field Theory

The first principle is locality. An action is an integral over spacetime of a density built from fields and their first derivatives at the same point:

$$S = \int d^4x \mathcal{L}(\phi, \phi_{,\mu}), \quad \phi_{,\mu} \equiv \partial_\mu \phi. \quad (1.28)$$

The comma is a compact derivative notation. The second principle is Lorentz invariance: $\mathcal{L}$ must be a scalar.

1.4.1 A scalar-field rehearsal

Before varying a four-vector, rehearse the method with one scalar field. A local Lorentz scalar may depend on a potential $U(\phi)$ and on the contraction of two derivatives:

$$\mathcal{L}_\phi = -\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - U(\phi). \quad (1.29)$$

With the mostly-plus metric,

$$\partial_\mu \phi \partial^\mu \phi = -\dot{\phi}^2 + |\nabla \phi|^2, \quad (1.30)$$

so

$$\mathcal{L}_\phi = \frac{1}{2} \dot{\phi}^2 - \frac{1}{2} |\nabla \phi|^2 - U(\phi). \quad (1.31)$$

The raised time derivative differs in sign from the lowered one; raised and lowered spatial derivatives do not.

Question from the audience. Does it mean anything to put the derivative index upstairs rather than downstairs?

Response. Yes. Raising the index contracts with the inverse metric. With signature $(-+++)$, $\partial^0 = -\partial_0$ while $\partial^i = \partial_i$. That is why the time-derivative term carries the opposite sign from the three spatial terms. ^a

^aLecture timestamp: 00:38:02.

The field Euler–Lagrange equation is

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \phi_{,\mu}} \right) = \frac{\partial \mathcal{L}}{\partial \phi}. \quad (1.32)$$

For (1.29),

$$\frac{\partial \mathcal{L}_\phi}{\partial \phi_{,\mu}} = -\partial^\mu \phi, \quad \frac{\partial \mathcal{L}_\phi}{\partial \phi} = -U'(\phi), \quad (1.33)$$

and hence

$$\square \phi = U'(\phi), \quad \square \equiv \partial_\mu \partial^\mu. \quad (1.34)$$

Equivalently,

$$\ddot{\phi} - \nabla^2 \phi = -U'(\phi). \quad (1.35)$$

On the board only the x -derivative is written before the y and z terms are left implicit. The result is the familiar wave equation with a force from the potential.

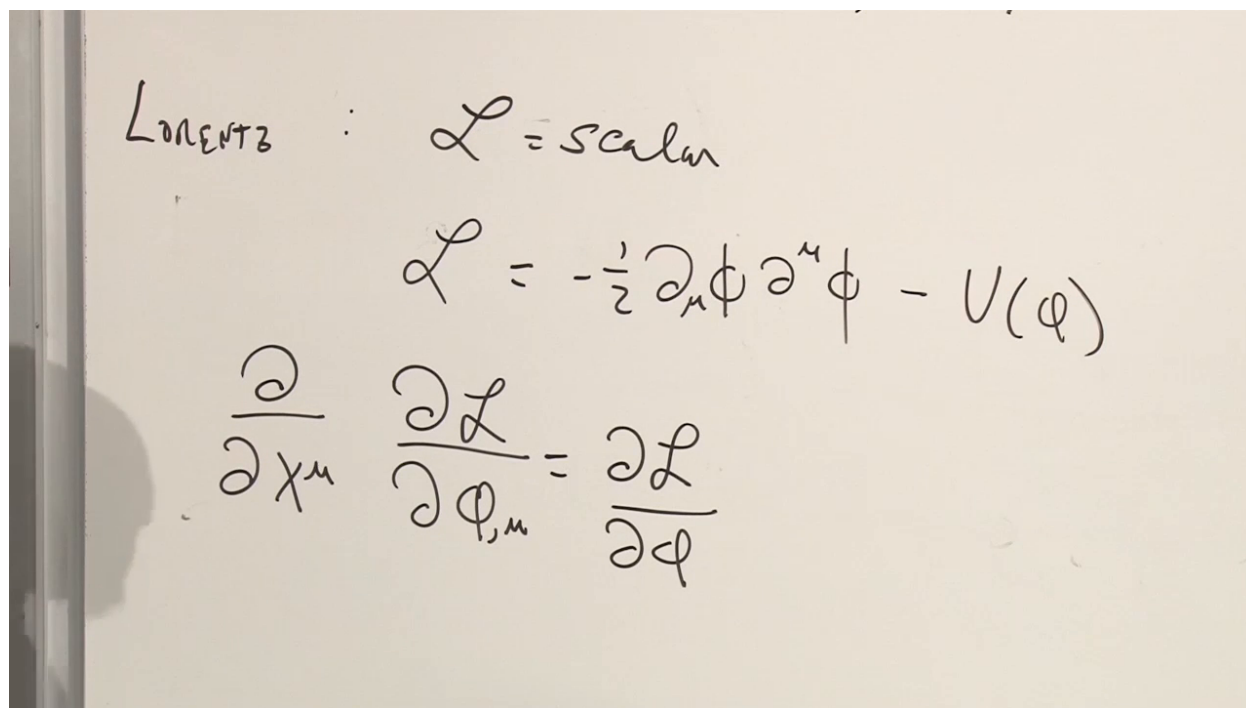


Figure 1.9: Locality, a scalar Lorentz-invariant Lagrangian, and the field-theory Euler–Lagrange equation.

The electromagnetic calculation is the same variational process with four fields and more indices.

1.5 Gauge Invariance Selects the Maxwell Scalar

Add a third principle: gauge invariance. For the moment set $J^\mu = 0$ and consider the free field. A gauge transformation is

$$A_\mu \longrightarrow A_\mu + \partial_\mu \chi. \quad (1.36)$$

The field tensor does not change because mixed second derivatives commute:

$$\begin{aligned} F_{\mu\nu} &\longrightarrow F_{\mu\nu} + \partial_\mu \partial_\nu \chi - \partial_\nu \partial_\mu \chi \\ &= F_{\mu\nu}. \end{aligned} \quad (1.37)$$

Lorentz invariance alone would allow the scalar

$$A_\mu A^\mu. \quad (1.38)$$

It is nevertheless forbidden in Maxwell theory because it changes under (1.36). The indices in this expression label field components; they are not derivatives. A derivative is indicated by a comma, as in $A_{\nu,\mu} = \partial_\mu A_\nu$.

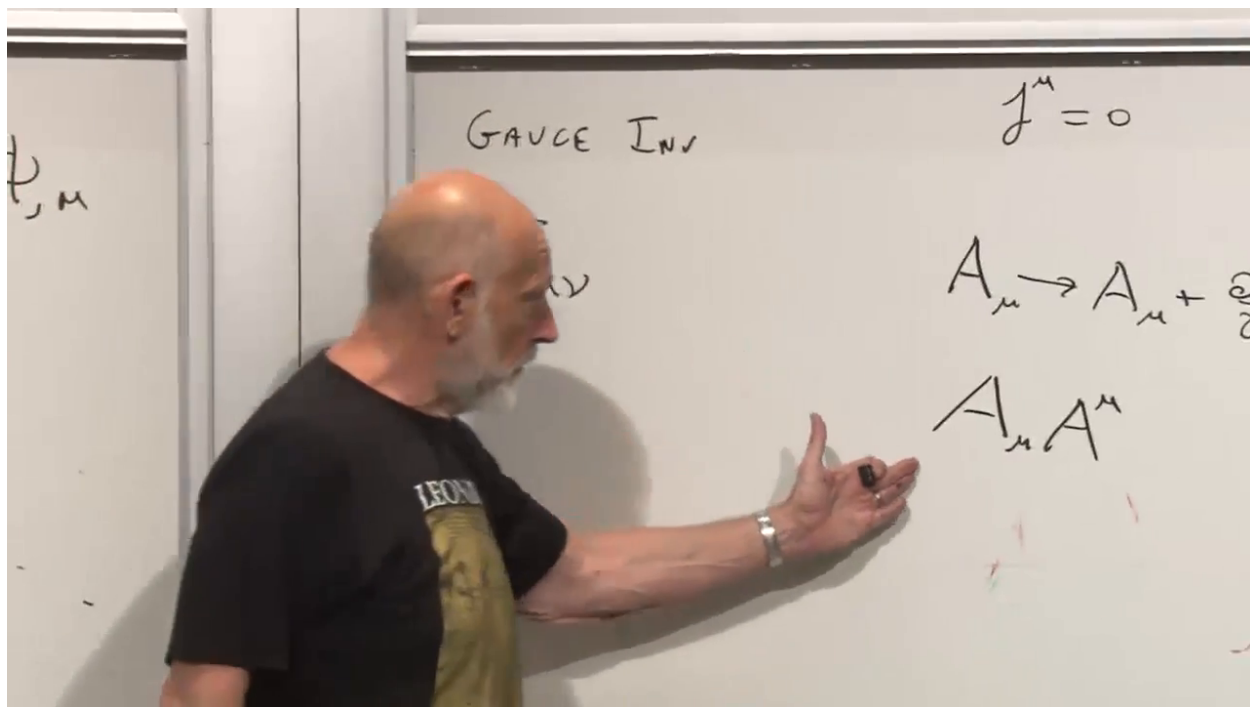


Figure 1.10: The gauge transformation and the Lorentz scalar $A_\mu A^\mu$, which is rejected because it is not gauge invariant.

Lecture frame: 00:43:40.

We should therefore build the free Lagrangian from $F_{\mu\nu}$. The linear trace fails:

$$F^\mu{}_\mu = 0. \quad (1.39)$$

Antisymmetry forces every diagonal component $F_{00}, F_{11}, F_{22}, F_{33}$ to vanish. Raising an index may change a sign, but it cannot turn zero into anything else.

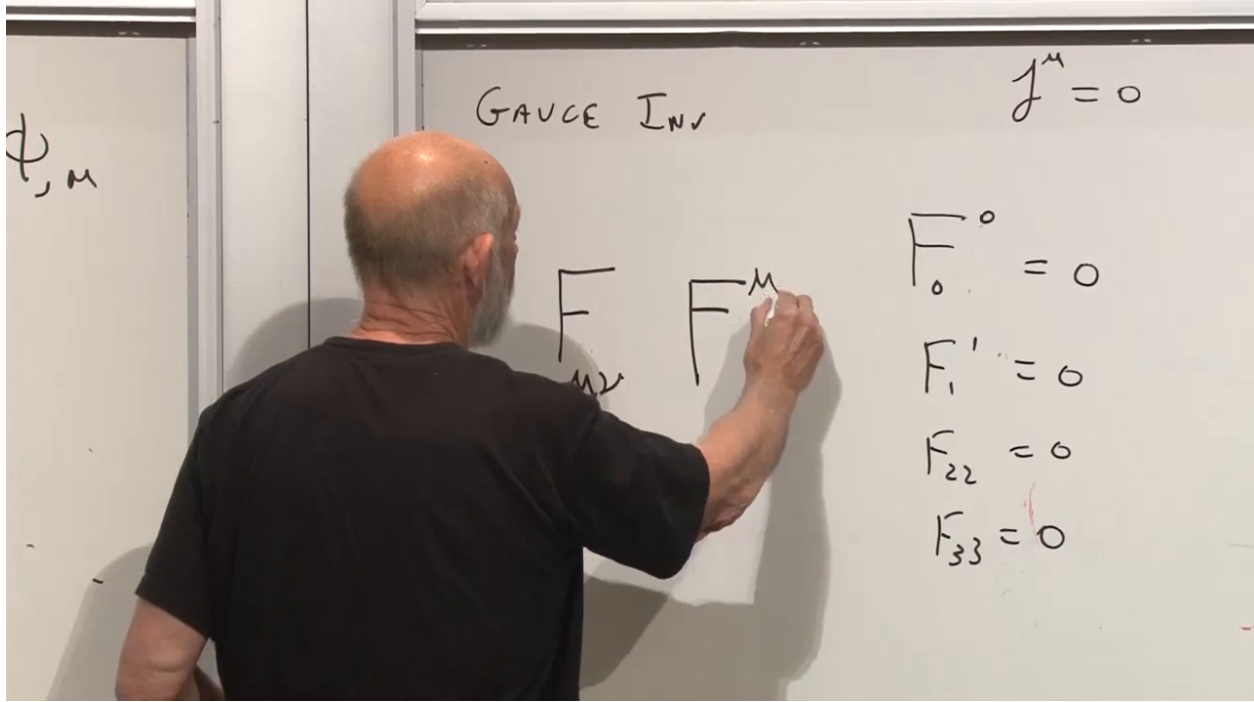


Figure 1.11: The attempted linear trace and the vanishing diagonal entries of the antisymmetric field tensor.

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The first nonzero Lorentz scalar is quadratic:

$$F_{\mu\nu}F^{\mu\nu}. \quad (1.40)$$

Let us unpack the contraction rather than quote it. The mixed components contain the electric field. Raising their time index contributes a minus sign, and both orderings $0i$ and $i0$ occur:

$$\sum_i (F_{0i}F^{0i} + F_{i0}F^{i0}) = -2E^2. \quad (1.41)$$

The purely spatial components contain the magnetic field. Raising spatial indices changes no sign, and both orderings again occur:

$$\sum_{i<j} (F_{ij}F^{ij} + F_{ji}F^{ji}) = 2B^2. \quad (1.42)$$

Therefore

$$F_{\mu\nu}F^{\mu\nu} = 2(B^2 - E^2), \quad -F_{\mu\nu}F^{\mu\nu} = 2(E^2 - B^2). \quad (1.43)$$

Figure 1.12: The completed contraction in terms of electric and magnetic field magnitudes.

Lecture frame: 00:51:00.

Question from the audience. Is F_{00} also zero?

Response. Yes. Every diagonal component of an antisymmetric tensor vanishes, including F_{00} . Only the off-diagonal mixed components, which encode $\mathbf{E}$, and the off-diagonal spatial components, which encode $\mathbf{B}$, contribute. ^a

^aLecture timestamp: 00:51:36.

The conventional normalization is

$$\mathcal{L}_{\text{EM}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} \quad (1.44)$$

or

$$\mathcal{L}_{\text{EM}} = \frac{1}{2}(E^2 - B^2). \quad (1.45)$$

The minus sign gives the electric term the familiar positive sign. The factor $1/4$ compensates the twofold appearance of each antisymmetric pair and leaves the conventional factor $1/2$. Multiplying the entire free action by a nonzero constant would not change its source-free equations, but this normalization is the one that fits the standard source coupling.

1.6 Varying the Free Electromagnetic Field

The independent fields are

$$A_0, \quad A_x, \quad A_y, \quad A_z. \quad (1.46)$$

For each component A_ν , the Euler–Lagrange equation is

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial A_{\nu,\mu}} \right) = \frac{\partial \mathcal{L}}{\partial A_\nu}, \quad A_{\nu,\mu} \equiv \partial_\mu A_\nu. \quad (1.47)$$

The placement in the comma notation is worth reading slowly: the first index labels which component of A is the field; the second labels the coordinate of differentiation.

In this notation,

$$F_{\mu\nu} = A_{\nu,\mu} - A_{\mu,\nu}. \quad (1.48)$$

This is exactly (1.4). Writing the comma indices in the opposite verbal order changes an intermediate minus sign, which is why a component calculation is useful.

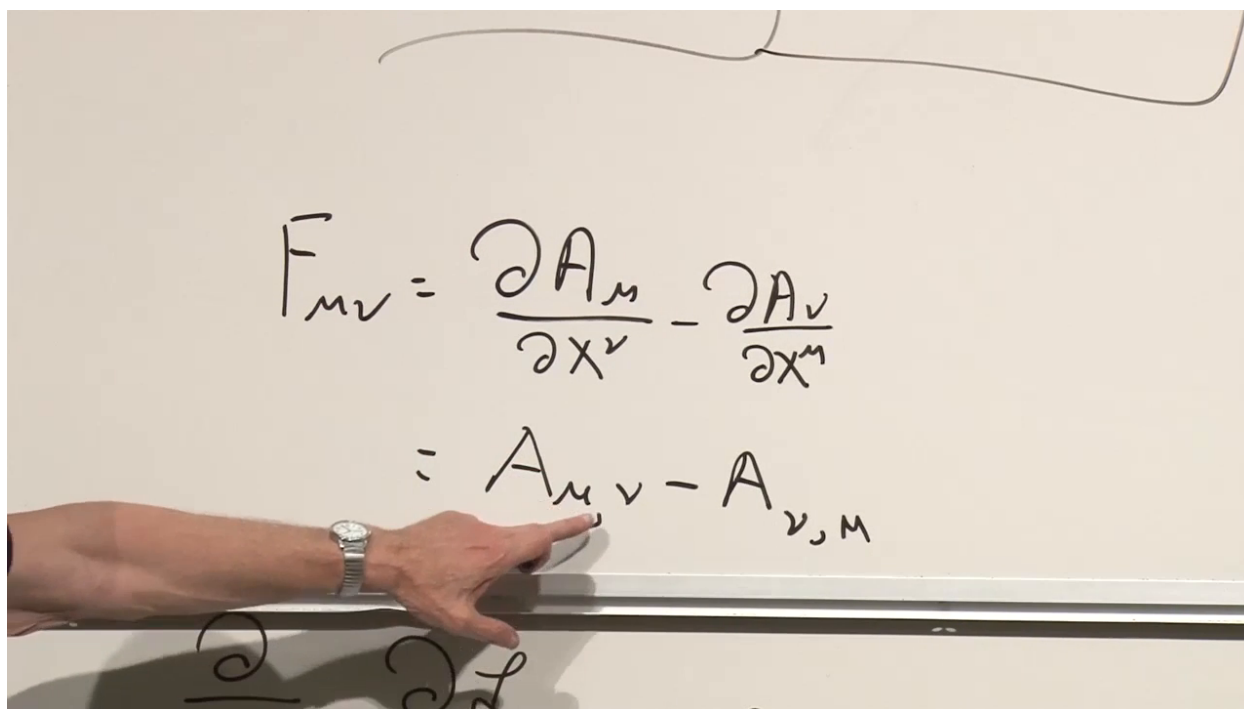


Figure 1.13: The field tensor expressed in comma notation after the four potential components have been identified as independent fields.

Lecture frame: 00:55:30.

Figure 1.14: The Maxwell Lagrangian written explicitly in terms of derivatives of the vector potential.

Lecture frame: 00:56:45.

1.6.1 One representative derivative

Select the derivative $A_{x,y} = \partial_y A_x$. The xy and yx terms in the full sum contribute

$$\begin{aligned}
 \mathcal{L}_{\text{EM}} &\supset -\frac{1}{4} (F_{xy}F^{xy} + F_{yx}F^{yx}) \\
 &= -\frac{1}{2} F_{xy}^2 \\
 &= -\frac{1}{2} (A_{x,y} - A_{y,x})^2.
 \end{aligned} \tag{1.49}$$

The last equality is safe because the sign disappears upon squaring: $A_{x,y} - A_{y,x} = F_{yx} = -F_{xy}$.

$$\mathcal{L} = -\frac{1}{4} (A_{m,\nu} - A_{\nu,m})(A^{m,\nu} - A^{\nu,m}) = -\frac{1}{2} \left(\frac{\partial A_x}{\partial y} - \frac{\partial A_y}{\partial x} \right)^2$$

$$\frac{\partial}{\partial x^4} \frac{\partial \mathcal{L}}{\partial A_{x,m}} \qquad \frac{\partial \mathcal{L}}{\partial A_{x,y}}$$

Figure 1.15: The representative $A_{x,y}$ derivative and the isolated antisymmetric square.*Lecture frame: 01:00:45.*

Differentiate:

$$\begin{aligned} \frac{\partial \mathcal{L}_{\text{EM}}}{\partial A_{x,y}} &= -(A_{x,y} - A_{y,x}) \\ &= F_{xy}. \end{aligned} \tag{1.50}$$

The board summarizes this with a minus sign attached to the tensor whose indices follow the board's derivative ordering. With the course-wide definition (1.4), the invariant statement is

$$\frac{\partial \mathcal{L}_{\text{EM}}}{\partial (\partial_\mu A_\nu)} = -F^{\mu\nu}. \tag{1.51}$$

For the example $\partial_y A_x$, this gives $-F^{yx} = F^{xy}$, precisely (1.50). Thus the apparent sign difference is only an index-order bookkeeping issue.

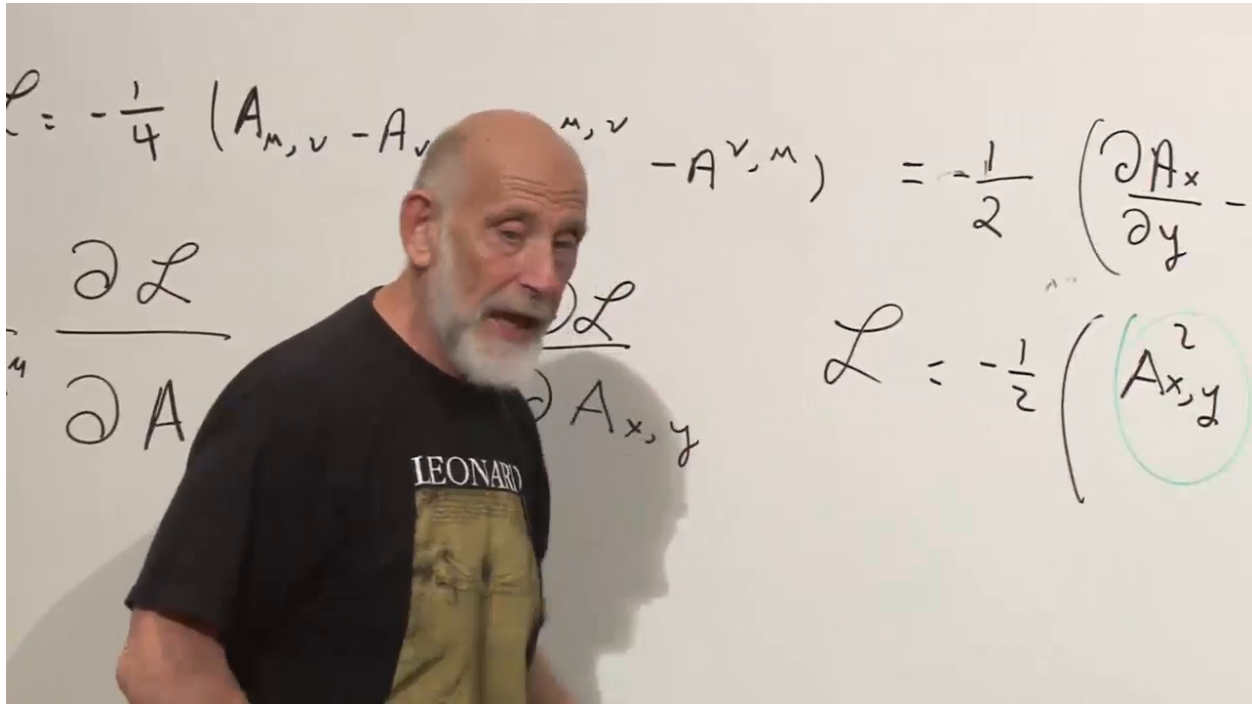


Figure 1.16: Expansion of the selected square before differentiating with respect to one derivative of the potential.

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Question from the audience. What happened to the factor of $1/2$ in the component derivative?

Response. Differentiating the square supplies a factor of two, which cancels the $1/2$:

$$\frac{d}{dx} \left[-\frac{1}{2} x^2 \right] = -x.$$

The earlier change from $1/4$ to $1/2$ came from the two ordered tensor components xy and yx .^a

^aLecture timestamp: 01:03:29.

There is no undifferentiated potential in the vacuum Lagrangian:

$$\frac{\partial \mathcal{L}_{\text{EM}}}{\partial A_\nu} = 0. \quad (1.52)$$

Substitution into (1.47) gives

$$-\partial_\mu F^{\mu\nu} = 0, \quad (1.53)$$

or equivalently

$$\boxed{\partial_\mu F^{\mu\nu} = 0.} \quad (1.54)$$

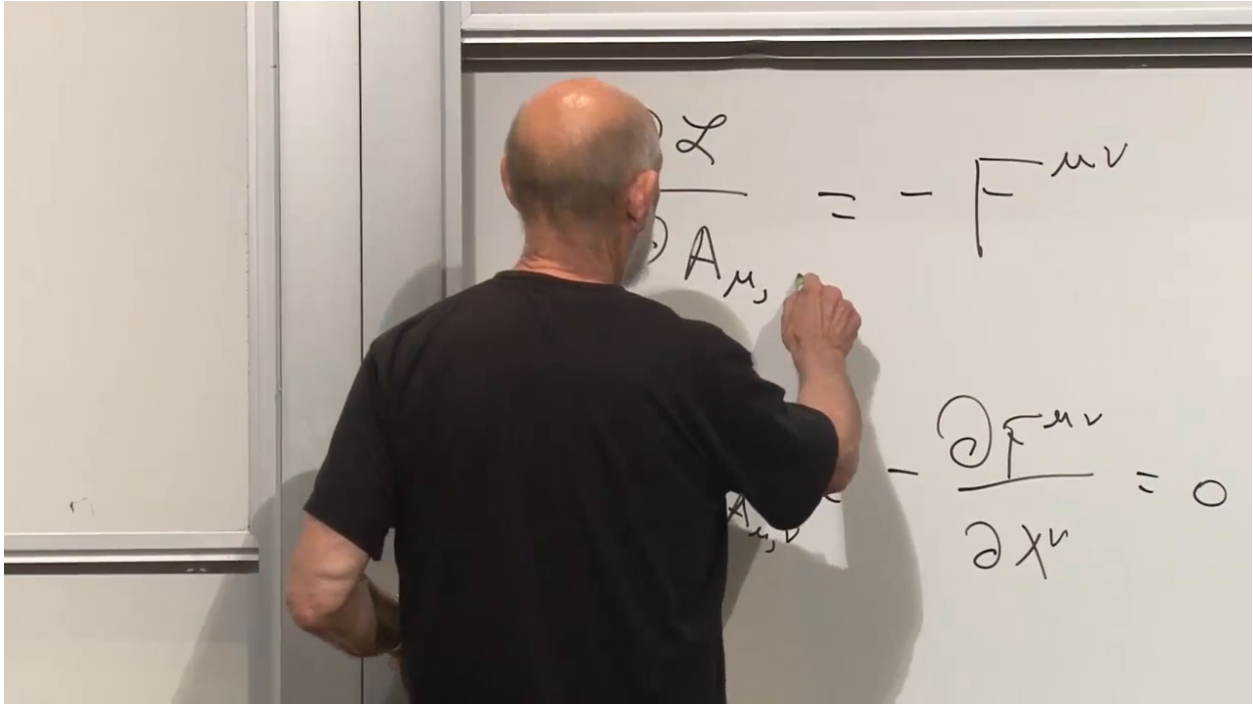


Figure 1.17: The component result generalized and inserted into the Euler–Lagrange equation.

Lecture frame: 01:09:00.

The same conclusion follows more cleanly by varying the whole action:

$$\begin{aligned}
 \delta S_{\text{EM}} &= -\frac{1}{2} \int d^4x F^{\mu\nu} (\partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu) \\
 &= - \int d^4x F^{\mu\nu} \partial_\mu \delta A_\nu \\
 &= \int d^4x (\partial_\mu F^{\mu\nu}) \delta A_\nu,
 \end{aligned} \tag{1.55}$$

where the boundary term is discarded. Antisymmetry combines the two terms in the first line. Since δA_ν is arbitrary, (1.54) follows.

Question from the audience. Did the isolated square assume that the magnetic field pointed only in the z -direction?

Response. No. It was one term pulled out of the full sum for inspection; an ellipsis belongs after it. The remaining spatial pairs give the other magnetic components, and the time–space pairs give the electric components. ^a

^aLecture timestamp: 01:08:05.

Question from the audience. Would choosing one time index instead of two spatial indices change the result?

Response. It gives the same general derivative with the metric sign handled by index raising. Two spatial indices select a magnetic component; one temporal and one spatial index select an electric component. The antisymmetric formula (1.51) covers both. ^a

^aLecture timestamp: 01:08:46.

Question from the audience. Are we still assuming $J^\mu = 0$?

Response. Only for this free-field derivation. The current was set to zero temporarily so that the gauge-invariant kinetic term could be identified. We now restore it and ask which extra term produces the right-hand side of Maxwell's equation. ^a

^aLecture timestamp: 01:09:41.

The algebra is not difficult, but it must be worked through rather than merely watched. Following a chain of indices once does not make it durable; reproducing the chain by hand does.

1.7 The Conserved Four-Current

The source is the four-current

$$J^\mu = (\rho, \mathbf{J}). \quad (1.56)$$

Its time component is charge density and its spatial components are the charge fluxes through three mutually perpendicular windows. Equivalently, for a small fixed box,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0. \quad (1.57)$$

Charge inside the box can decrease only by flowing through its boundary. In covariant notation,

$$\boxed{\partial_\mu J^\mu = 0.} \quad (1.58)$$

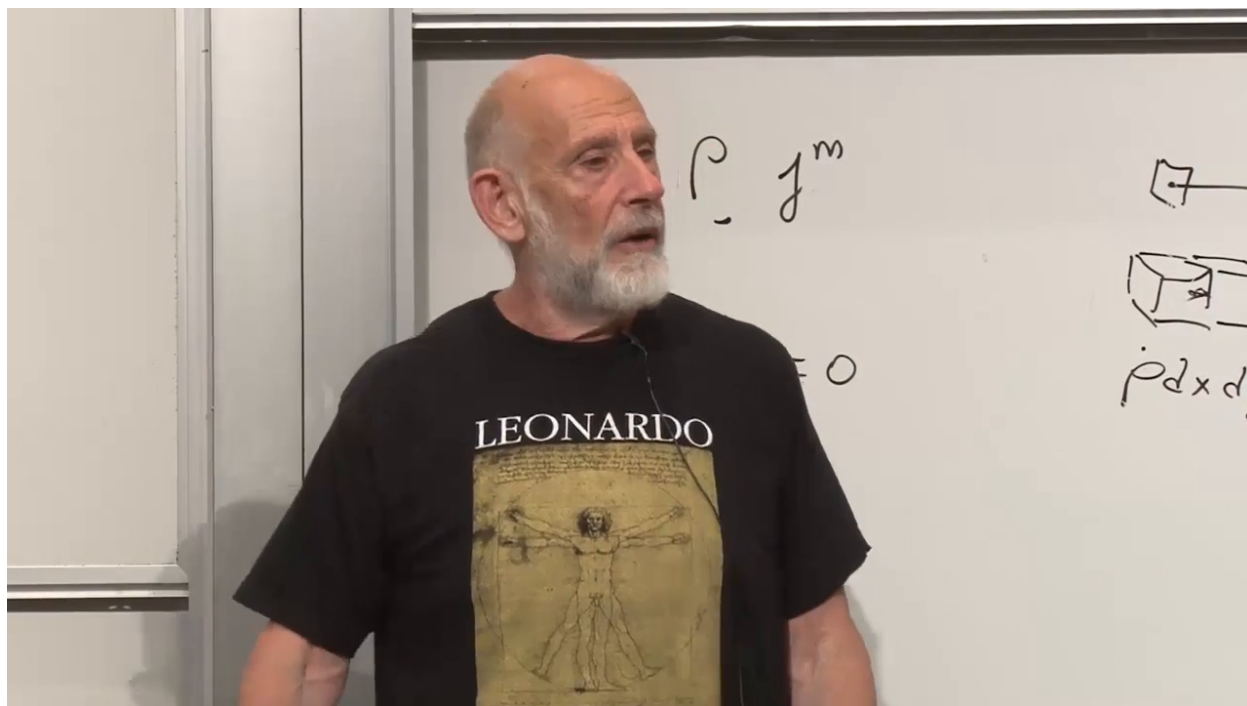


Figure 1.18: Charge density, spatial current, and the small-box interpretation recalled before the source coupling is introduced.

Lecture frame: 01:14:45.

1.8 The Source Coupling and Gauge Invariance

The simplest local Lorentz scalar linear in both the current and the potential is

$$S_{\text{int}} = - \int d^4x J^\mu A_\mu. \quad (1.59)$$

The overall sign is conventional. The lecture jokes that the historical choice of electric-charge sign can ultimately be blamed on Benjamin Franklin; the real point is that one consistent sign convention must be used everywhere.

At first (1.59) does not look gauge invariant. Under (1.36),

$$\delta_\chi S_{\text{int}} = - \int d^4x J^\mu \partial_\mu \chi. \quad (1.60)$$

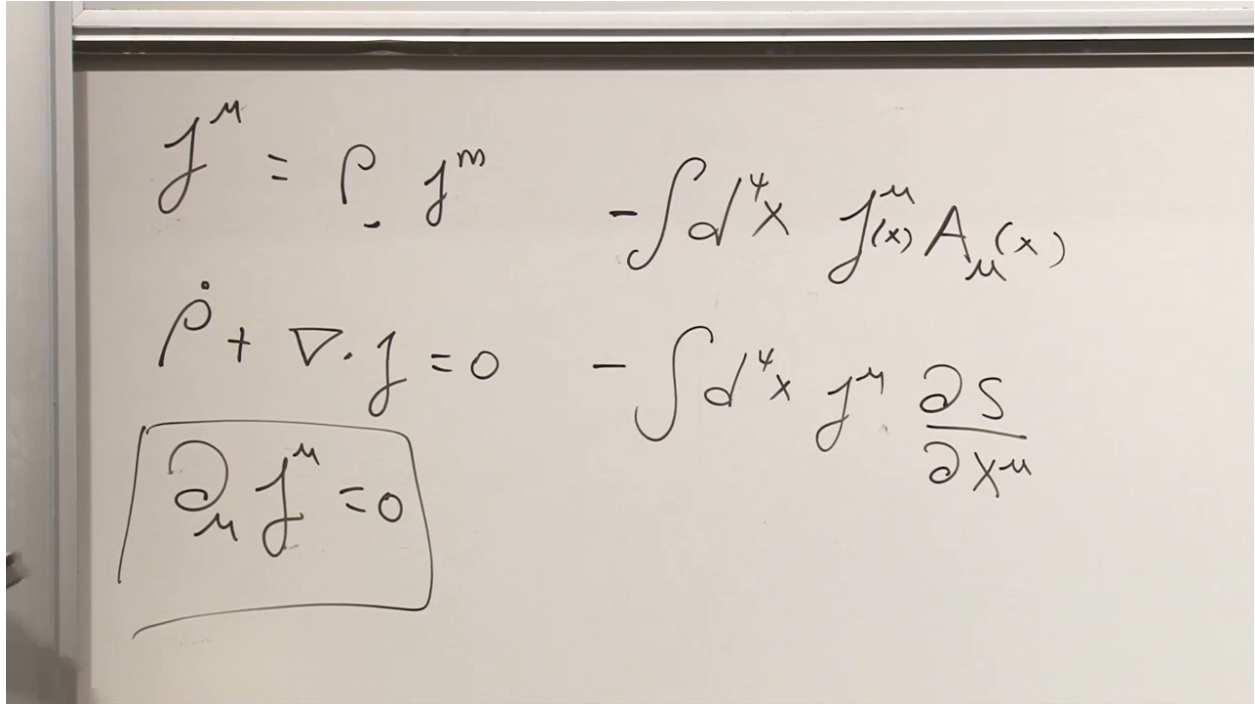


Figure 1.19: The interaction action and its variation under $A_\mu \mapsto A_\mu + \partial_\mu \chi$.

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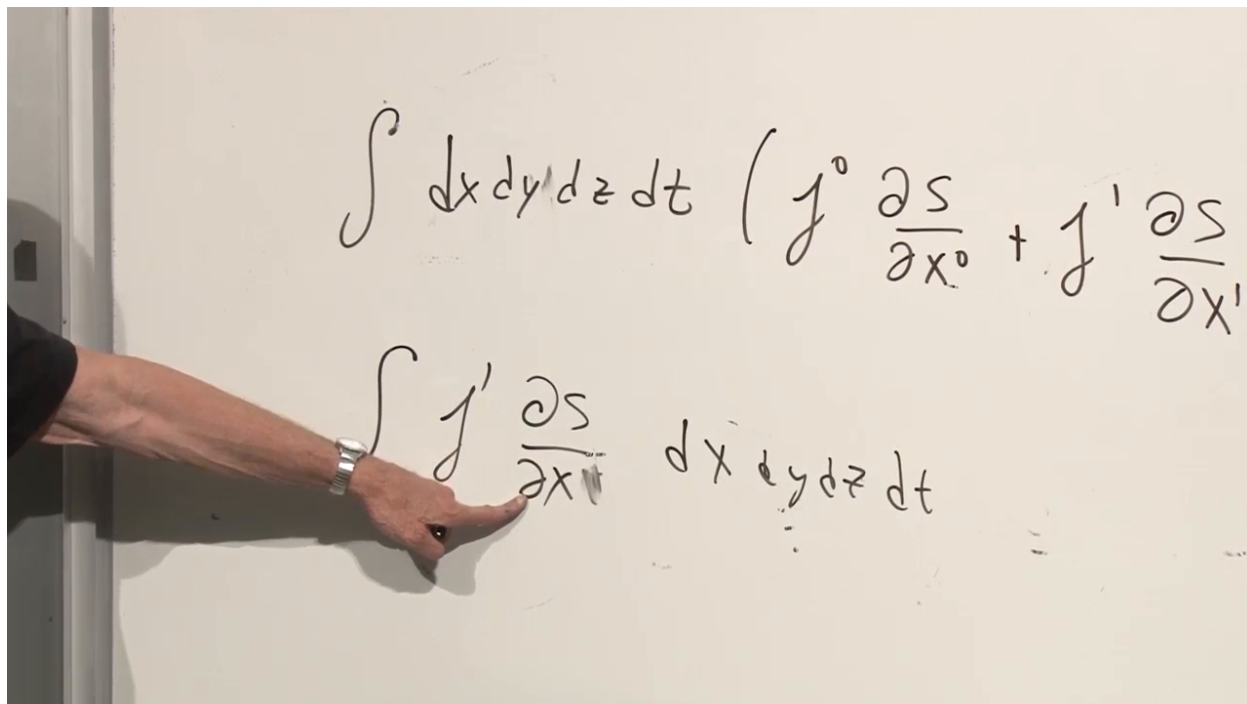
To see the mechanism in elementary terms, expand the contraction:

$$J^\mu \partial_\mu \chi = J^0 \partial_0 \chi + J^1 \partial_1 \chi + J^2 \partial_2 \chi + J^3 \partial_3 \chi. \quad (1.61)$$

Assume the current is confined to a large laboratory or falls off fast enough that it vanishes at the boundary. Integrate each term by parts. For the first spatial coordinate, for example,

$$\begin{aligned} - \int d^4x J^1 \partial_1 \chi &= - \int dt dx^2 dx^3 \left[J^1 \chi \right]_{x^1=-\infty}^{x^1=+\infty} \\ &\quad + \int d^4x \chi \partial_1 J^1 \\ &= \int d^4x \chi \partial_1 J^1. \end{aligned} \quad (1.62)$$

The same operation applies to the time term and the other two spatial terms.



$$\int dx dy dz dt \left(\gamma^0 \frac{\partial S}{\partial x^0} + \gamma^1 \frac{\partial S}{\partial x^1} \right)$$

$$\int \gamma^1 \frac{\partial S}{\partial x^1} dx dy dz dt$$

Figure 1.20: The four gauge-variation terms expanded before integration by parts.

Lecture frame: 01:19:15.

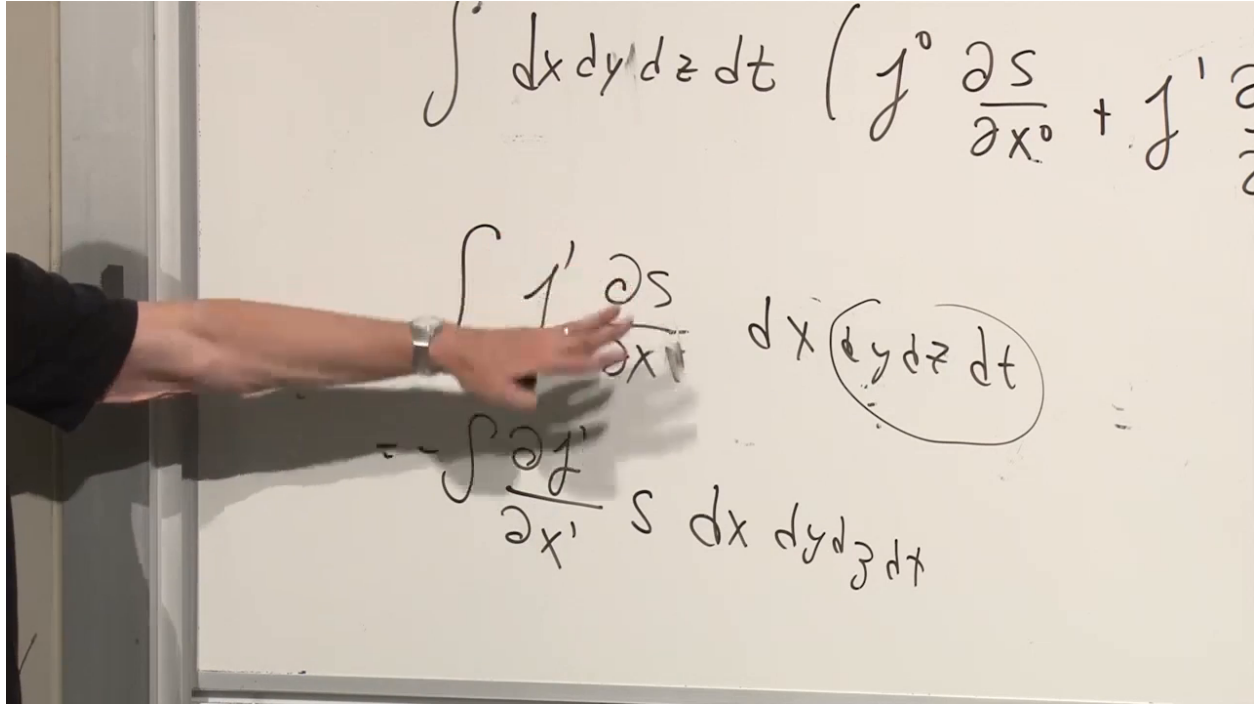


Figure 1.21: A representative integration by parts moves the derivative from the gauge function to the current.

Lecture frame: 01:20:45.

Collecting all four components,

$$\delta_\chi S_{\text{int}} = \int d^4x \chi \partial_\mu J^\mu. \quad (1.63)$$

Consequently,

$$\partial_\mu J^\mu = 0 \implies \delta_\chi S_{\text{int}} = 0. \quad (1.64)$$

Within the assumed boundary conditions, the source interaction is gauge invariant precisely when the current is locally conserved.

Question from the audience. Is it enough to require only $\nabla \cdot \mathbf{J} = 0$?

Response. No. All four terms are required:

$$\partial_\mu J^\mu = \dot{\rho} + \nabla \cdot \mathbf{J} = 0.$$

The first term is the change of charge density in the fixed box. A nonzero spatial divergence is permitted when it is balanced by that change. ^a

^aLecture timestamp: 01:22:00.

1.9 The Sourced Maxwell Equation

The complete electromagnetic Lagrangian density is

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - J^\mu A_\mu. \quad (1.65)$$

The source term contains no derivative of A_ν , so it leaves (1.51) unchanged and contributes

$$\frac{\partial \mathcal{L}}{\partial A_\nu} = -J^\nu. \quad (1.66)$$

The Euler–Lagrange equation therefore gives

$$-\partial_\mu F^{\mu\nu} = -J^\nu, \quad (1.67)$$

or

$$\boxed{\partial_\mu F^{\mu\nu} = J^\nu.} \quad (1.68)$$

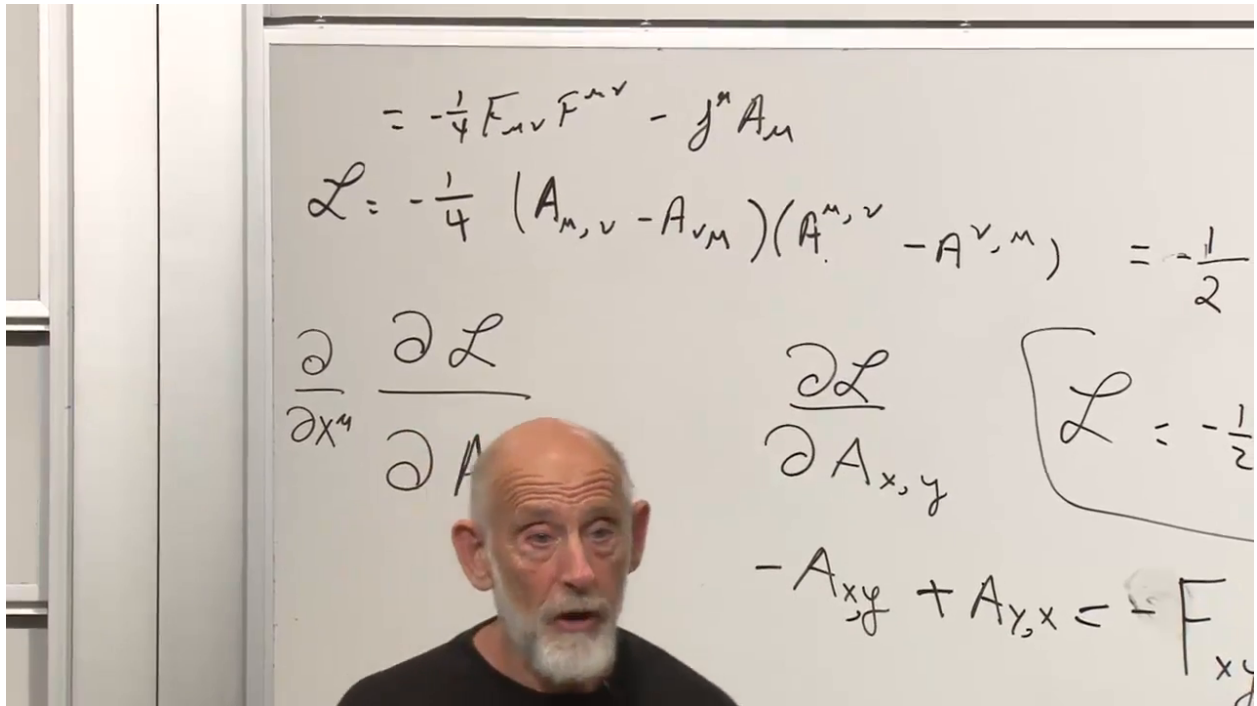


Figure 1.22: The free and source terms displayed with the representative Euler–Lagrange derivatives.

Lecture frame: 01:25:00.

A nonconserved prescribed current would cause two failures at once. The interaction action would not be gauge invariant, and the sourced Maxwell equations would be inconsistent. Charge cannot simply disappear inside a region without a compensating flux through its boundary.

1.10 Maxwell's Equations Give Conservation Back

The compatibility can be read directly from the field equation. Taking ∂_ν of (1.68) yields

$$\partial_\nu J^\nu = \partial_\nu \partial_\mu F^{\mu\nu} = 0, \quad (1.69)$$

because the derivatives are symmetric under $\mu \leftrightarrow \nu$ while $F^{\mu\nu}$ is antisymmetric.

The lecture also checks this in three-vector form, and the check exposes a live sign error. Start with Gauss's law:

$$\rho = \nabla \cdot \mathbf{E}. \quad (1.70)$$

Differentiate in time:

$$\dot{\rho} = \nabla \cdot \dot{\mathbf{E}}. \quad (1.71)$$

The provisional board equation was written

$$\dot{\mathbf{E}} = -\nabla \times \mathbf{B} + \mathbf{J}. \quad (1.72)$$

Its divergence would give

$$\dot{\rho} = \nabla \cdot \mathbf{J}, \quad (1.73)$$

since the divergence of a curl vanishes. That contradicts (1.57).

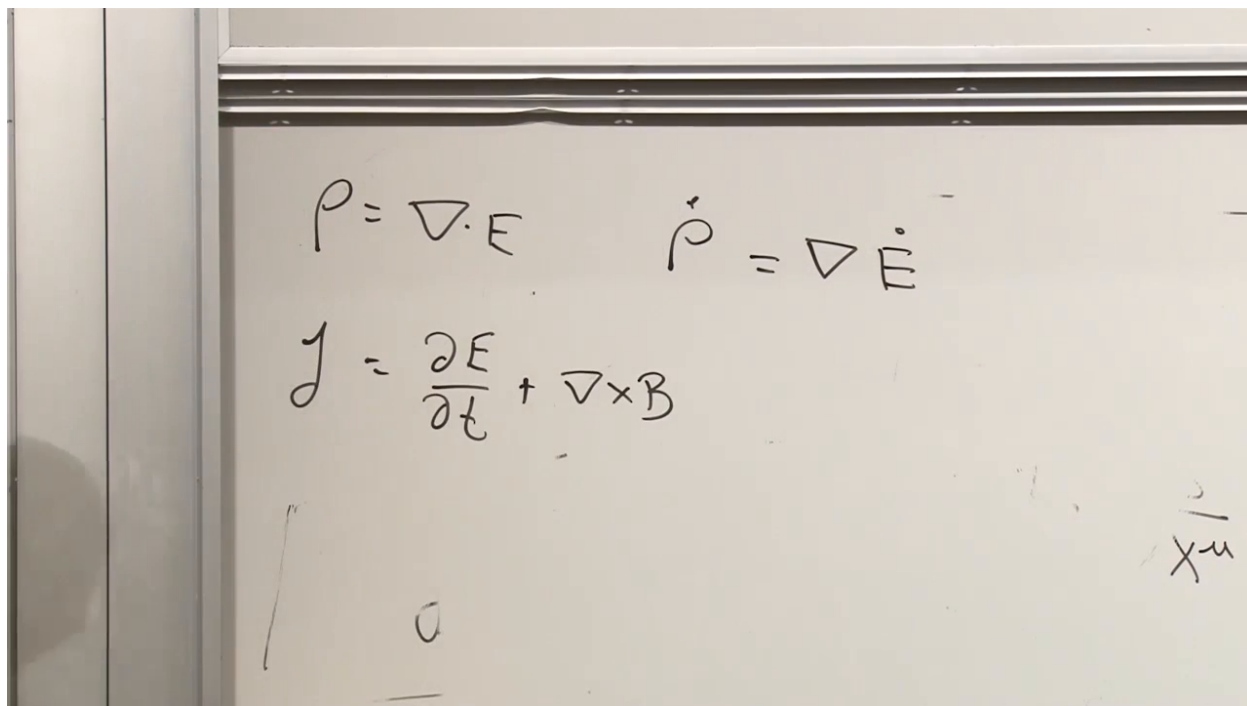


Figure 1.23: The provisional plus- $\mathbf{J}$ board equation that produces the wrong continuity sign.

Lecture frame: 01:27:00.

The mismatch is not hidden. The calculation is retraced in real time; the required continuity equation is stated; and a member of the audience checks the equation and identifies the source sign. The conventionally consistent component equation is

$$\boxed{\dot{\mathbf{E}} + \nabla \times \mathbf{B} = -\mathbf{J}}, \quad (1.74)$$

or

$$\dot{\mathbf{E}} = -\nabla \times \mathbf{B} - \mathbf{J}. \quad (1.75)$$

Now equations (1.71) and (1.74) give

$$\begin{aligned} \dot{\rho} &= \nabla \cdot (-\nabla \times \mathbf{B} - \mathbf{J}) \\ &= -\nabla \cdot \mathbf{J}, \end{aligned} \quad (1.76)$$

which is exactly local charge conservation.

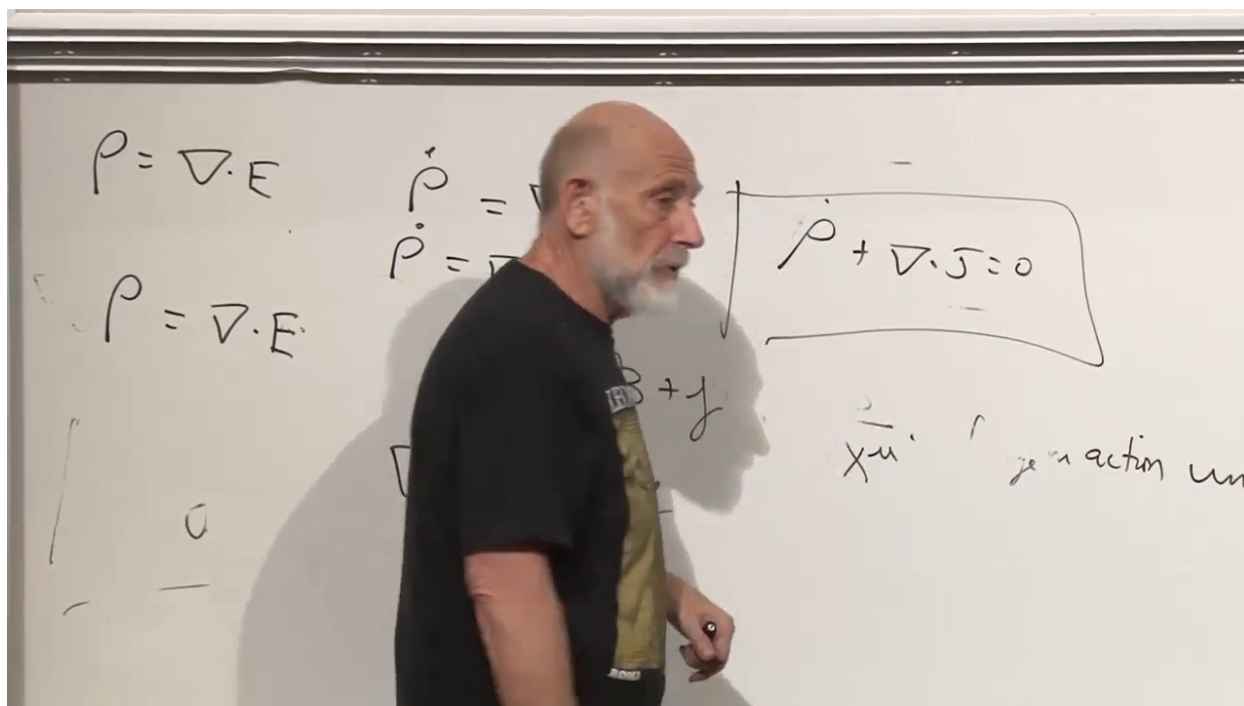


Figure 1.24: The corrected source sign and the recovered continuity equation.

Lecture frame: 01:31:00.

Question from the audience. Should the current term on the right be $-\mathbf{J}$?

Response. Yes. With the field-tensor convention used throughout the course, the spatial part of $\partial_\mu F^{\mu\nu} = J^\nu$ is $\dot{\mathbf{E}} + \nabla \times \mathbf{B} = -\mathbf{J}$. That correction makes the divergence calculation reproduce $\dot{\rho} + \nabla \cdot \mathbf{J} = 0$ and does not alter the preceding covariant action derivation. ^a

^aLecture timestamp: 01:30:17.

1.11 Which Principle Is Fundamental?

Question from the audience. Is the reasoning circular? We used charge conservation to make the action gauge invariant, then used the action to derive Maxwell's equations, and Maxwell's equations imply charge conservation.

Response. The statements are mutually compatible, not a circular proof that assumes its own conclusion. One may begin with observed Maxwell equations and discover the continuity equation; then $-J^\mu A_\mu$ is recognized as a gauge-invariant coupling. Or one may begin with locality, Lorentz invariance, gauge invariance, and an action; gauge invariance requires a conserved current and the action yields Maxwell's equations. Or one may regard the physical fact that charge does not disappear locally as primary. Each route reaches the same structure. ^a

^aLecture timestamp: 01:31:11.

If one asks for the most economical constructive route, the combination of locality, Lorentz invariance,

gauge invariance, and a Lagrangian principle selects

$$-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - J^\mu A_\mu \quad (1.77)$$

and hence Maxwell's dynamical equation. Calling one entry in this triangle more fundamental is partly a matter of taste. The important fact is that the entries fit.

1.12 Why the Word *Gauge*?

Question from the audience. The mathematics is clear, but what does the word *gauge* mean here?

Response. The name is a historical anachronism. A gauge can refer to a chosen zero, as in pressure measured relative to a reference. Hermann Weyl originally connected the idea to a local standard of length; that interpretation was not the right physics. The term survived when the freedom was correctly connected to the phase of a quantum wave function. In the present classical calculation, changing A_μ by $\partial_\mu\chi$ changes the descriptive zero without changing $F_{\mu\nu}$. The deeper quantum meaning is deferred. ^a

^aLecture timestamp: 01:34:49.

The historical label should not obscure the operative statement: potentials related by (1.36) describe the same electromagnetic field. Gauge freedom is therefore a redundancy of the potential description, while the field tensor is gauge invariant.

1.13 Why the Two Maxwell Families Look Asymmetric

Question from the audience. If electric and magnetic fields are so symmetric, why does one Maxwell family appear as a Bianchi identity while the other comes from equations of motion?

Response. In source-free electromagnetism one may introduce a dual potential so that the roles are exchanged: what was an equation of motion becomes a Bianchi identity and vice versa. The physical asymmetry enters through sources. Electric charges are observed, while elementary magnetic monopoles have not been observed. If there were an exact ordinary symmetry exchanging electric and magnetic charges, one would expect magnetic counterparts with corresponding spectra, which are not seen. More sophisticated theories can still contain magnetic charges or electric–magnetic dualities, but that is beyond the present derivation. ^a

^aLecture timestamp: 01:37:11.

Thus the source-free equations possess an electric–magnetic interchange structure, while the observed source sector chooses an electric description. That is more than a typographical convention, but it does not invalidate the symmetry visible in a free plane wave.

1.14 What Has Been Assembled

The route through the lecture is now complete:

1. Maxwell's equations force a plane wave to be transverse, with orthogonal electric and magnetic fields and dispersion $\omega = ck$.
2. The Bianchi equations are identities following from the potential; the contracted equations are dynamical.
3. Locality and Lorentz invariance lead us to an action density, while gauge invariance rules out $A_\mu A^\mu$ and selects a scalar made from $F_{\mu\nu}$.
4. The free action $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ yields the vacuum Maxwell equation.
5. The interaction $-J^\mu A_\mu$ is gauge invariant exactly when the current obeys the continuity equation.
6. The sourced action yields $\partial_\mu F^{\mu\nu} = J^\nu$, and antisymmetry gives $\partial_\mu J^\mu = 0$ back.

This is a large amount of structure: Lagrangian mechanics, relativistic field theory, electrodynamics, Lorentz invariance, gauge invariance, and local charge conservation have become parts of one argument. The only reliable way to digest it is to start again with the equations and carry each derivative and sign through by hand.